CLOUD-BASED FILE STORAGE SYSTEM

OMAR CUSMA FAIT - CLOUD COMPUTING BASIC COURSE – 2023/2024

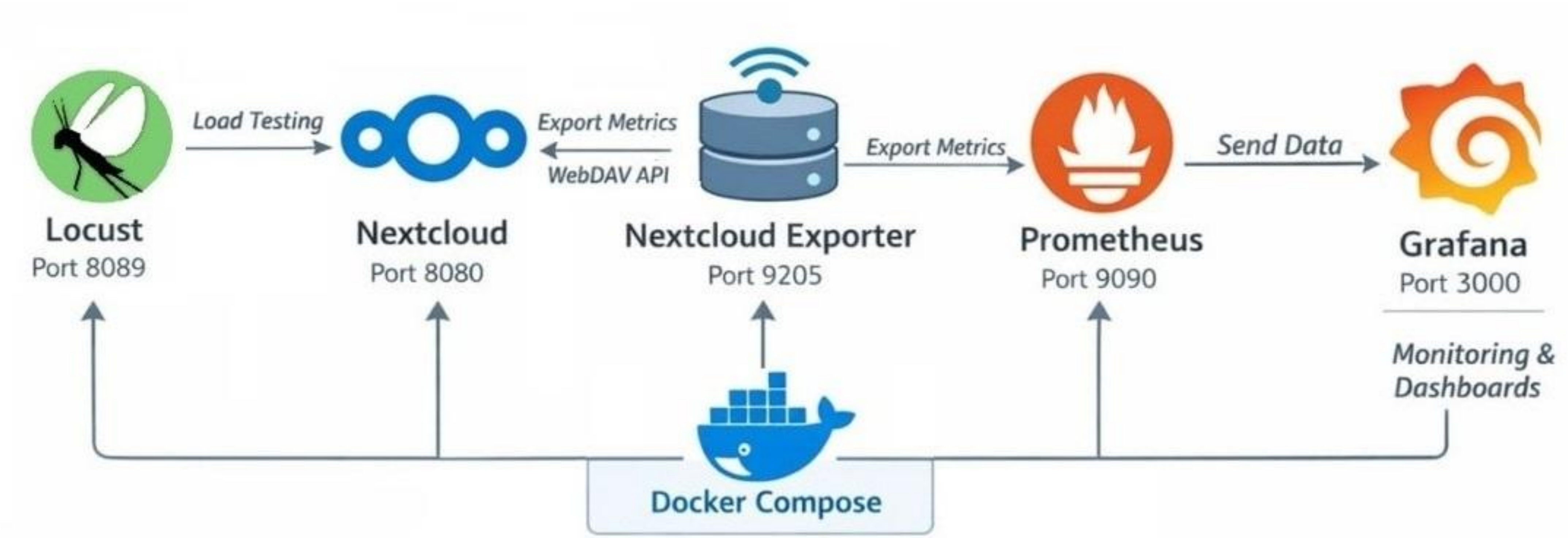
PROJECT OVERVIEW

Objective:
Deploy a self-hosted cloud storage platform with monitoring and performance testing capabilities

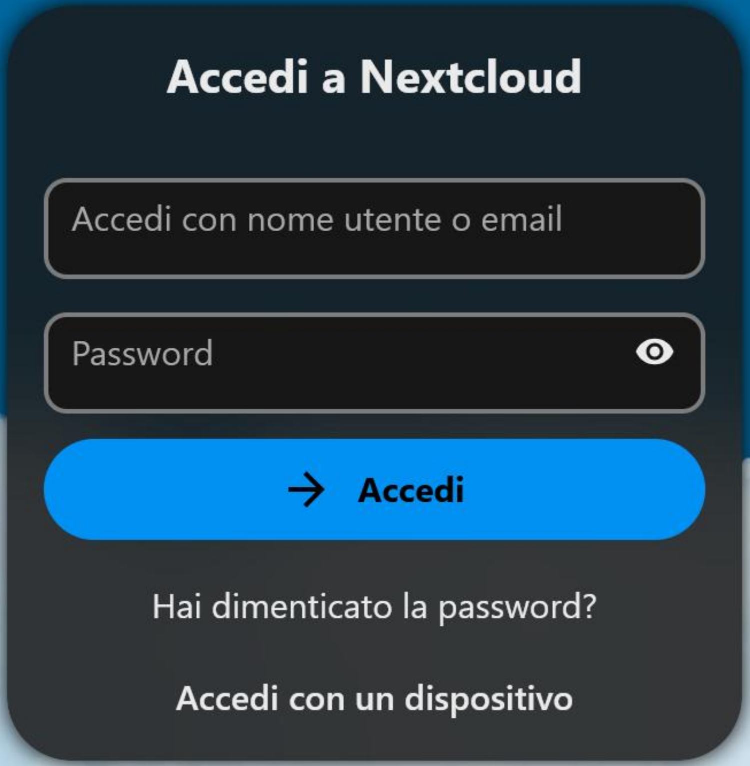
Requirements Addressed:
✓ User authentication & authorization
✓ Private storage spaces
✓ File operations (upload/download/delete)
✓ Scalability, security, and cost-efficiency

Key Components:
- **Nextcloud**: File storage and user management
- **Docker Compose**: Container orchestration
- **Prometheus + Grafana**: Monitoring and visualization
- **Locust**: Load testing and performance analysis

SYSTEM ARCHITECTURE



NEXTCLOUD FEATURES



Admin Capabilities:

- ✓ User account creation and management
- ✓ Private storage allocation per user (`/remote.php/dav/files/username/`)
- ✓ Group management and permissions
- ✓ System configuration and security settings

User Capabilities:

- ✓ Upload/download/delete files
- ✓ Sign-up, log out
- ✓ File preview (images, PDFs, text)
- ✓ Folder creation and organization
- ✓ Web interface and desktop sync

To log in, visit http://localhost:8080

Nextcloud – un posto sicuro per tutti i tuoi dati

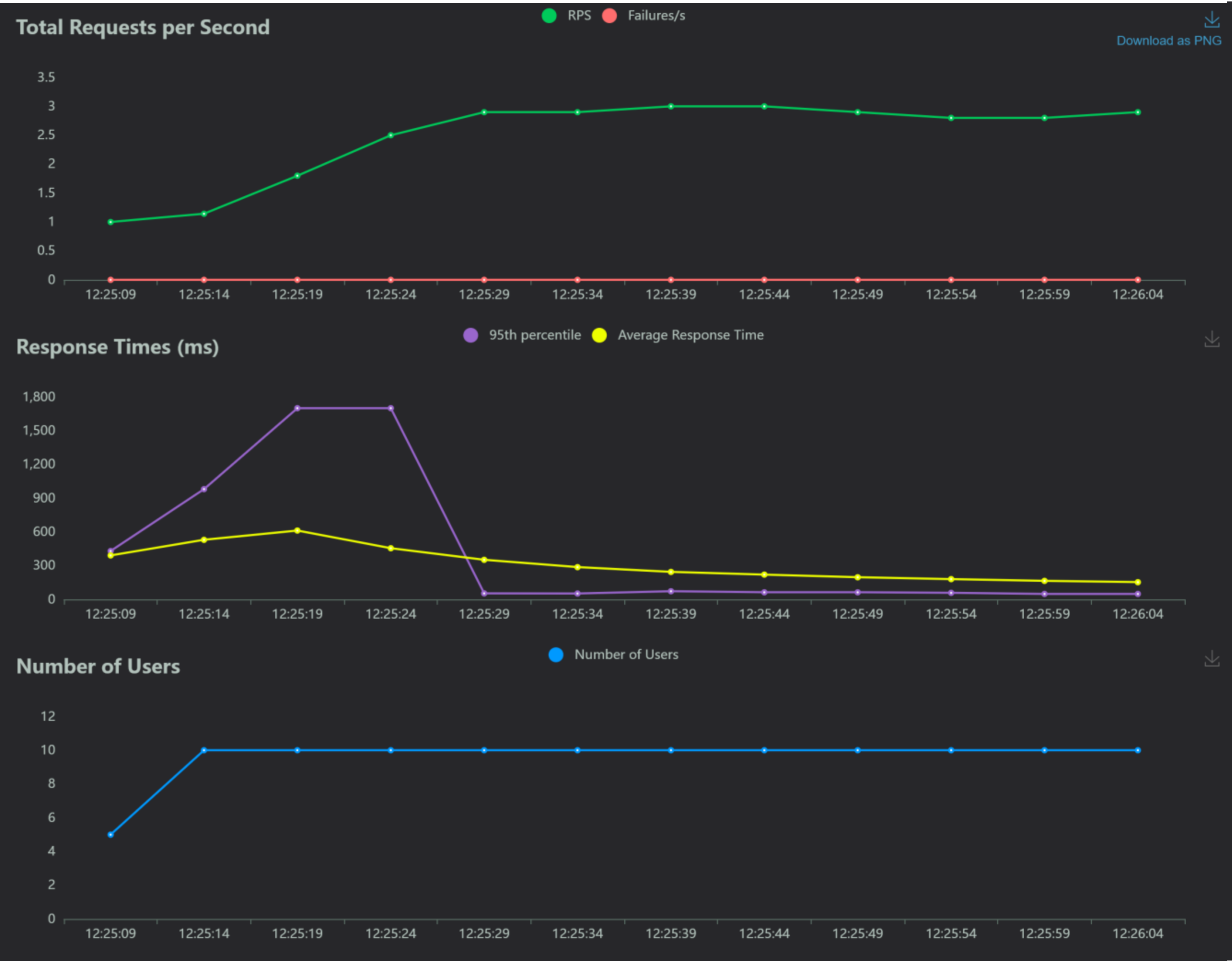
MONITORING STACK

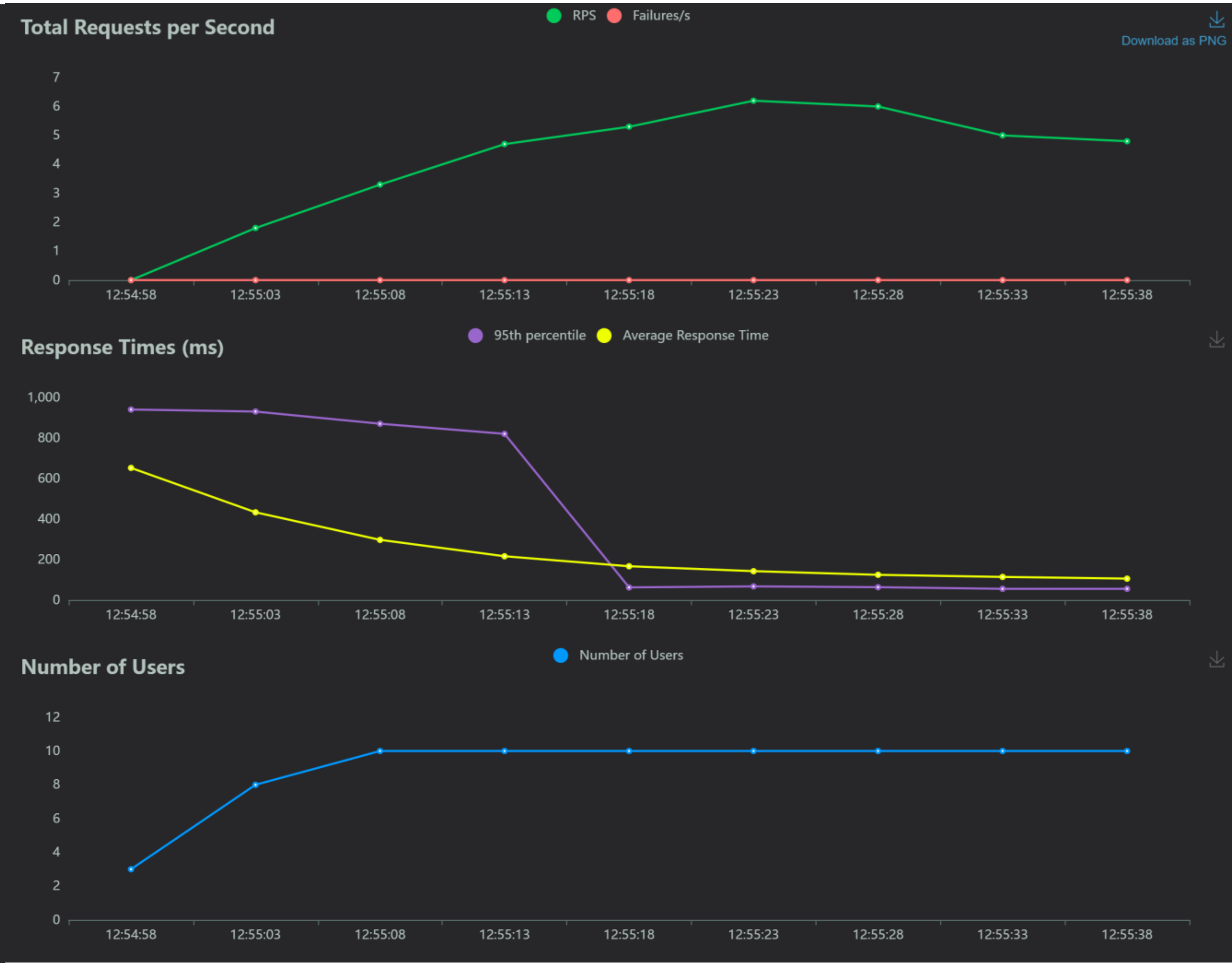
Nextcloud Exporter
- Scrapes Nextcloud internal metrics
- Exposes data at `*http://localhost:9205/metrics*`
- Key metrics: active users, file count, response times

Prometheus
- Time-series database
- Scrape interval: 30 seconds
- Configuration: `prometheus/prometheus.yml`

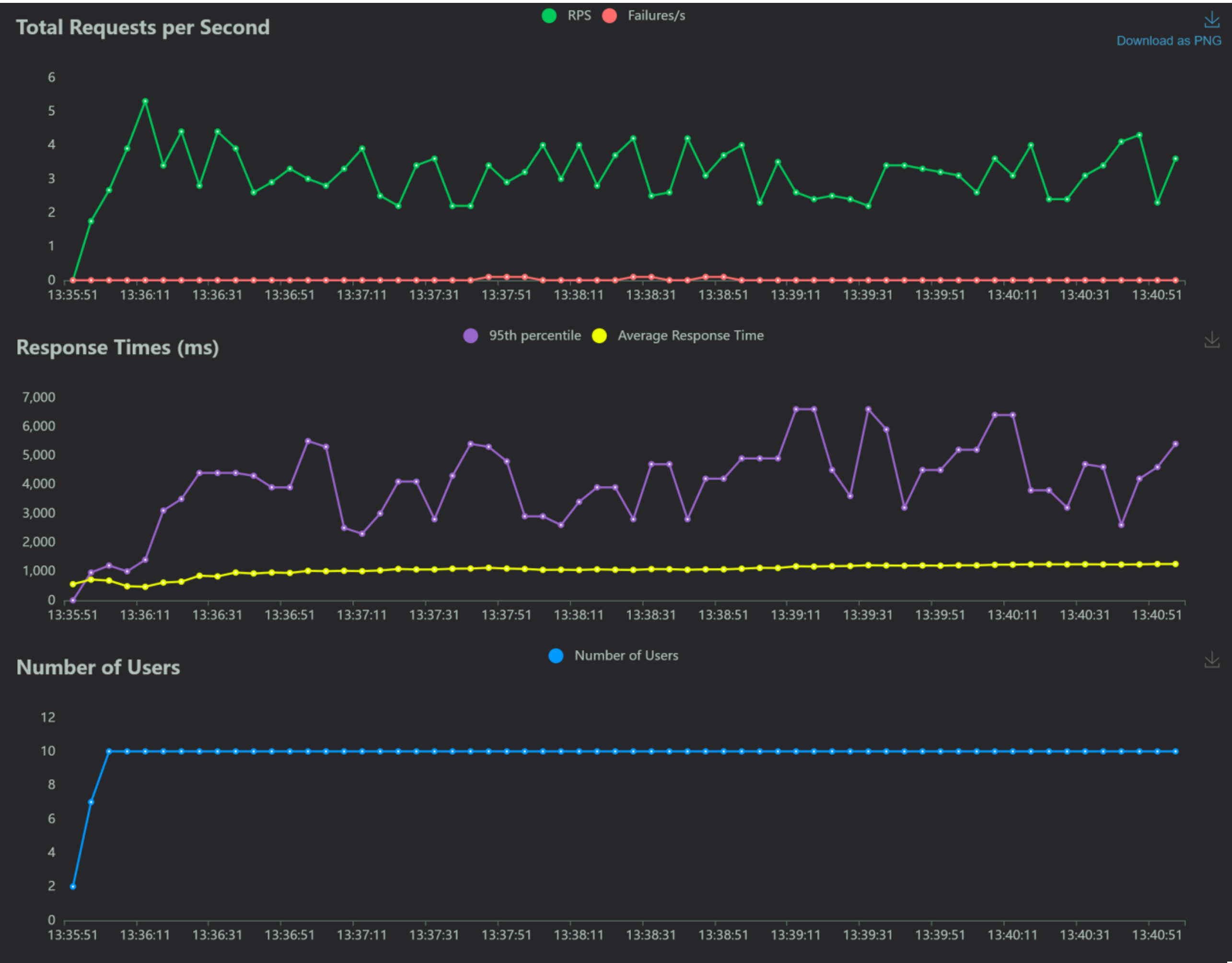
Grafana
- Auto-provisioned Prometheus datasource
- Custom dashboards for real-time monitoring
- Access: `*http://localhost:3000*`

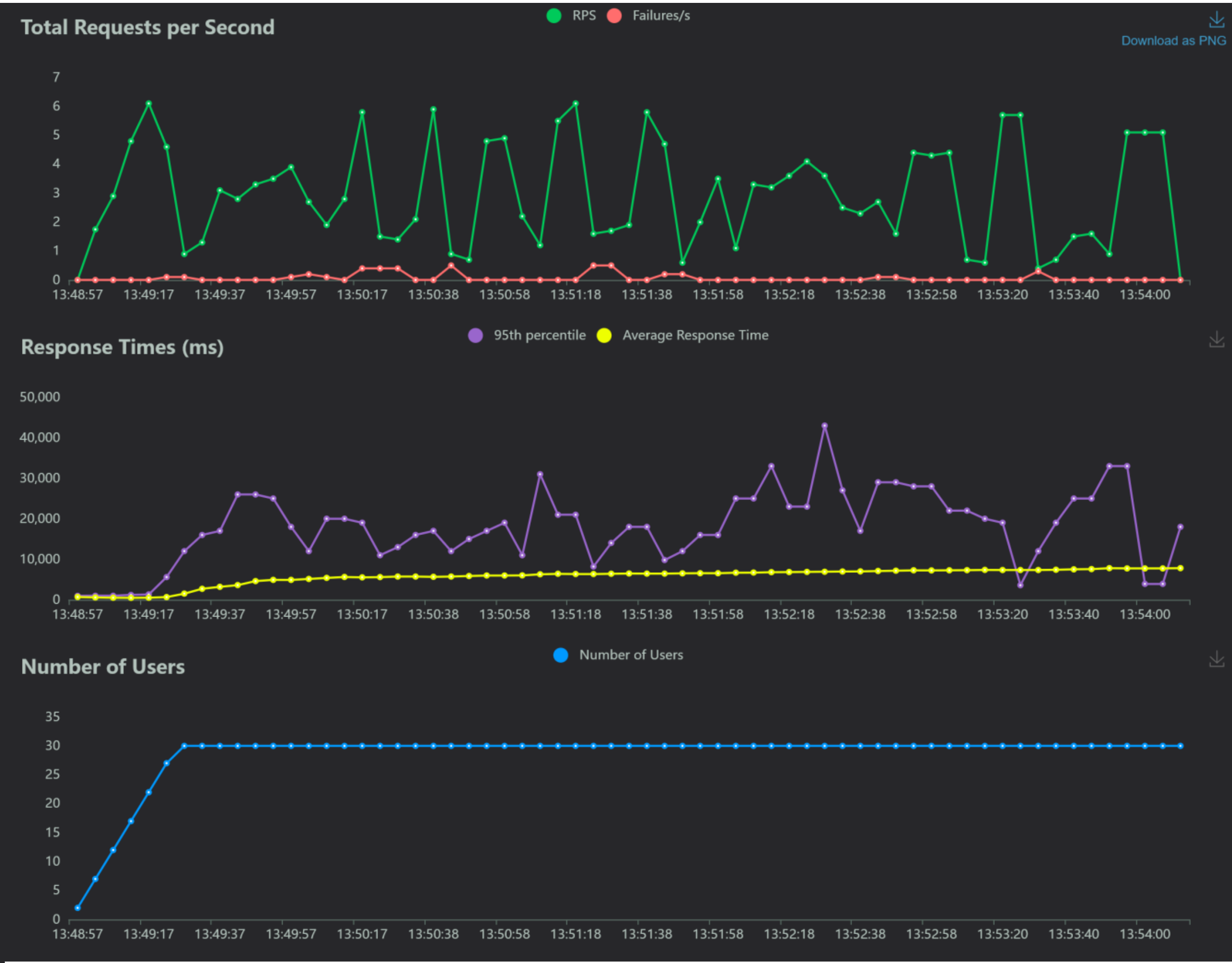
LOAD-TESTING RESULTS: 500KB - 5MB





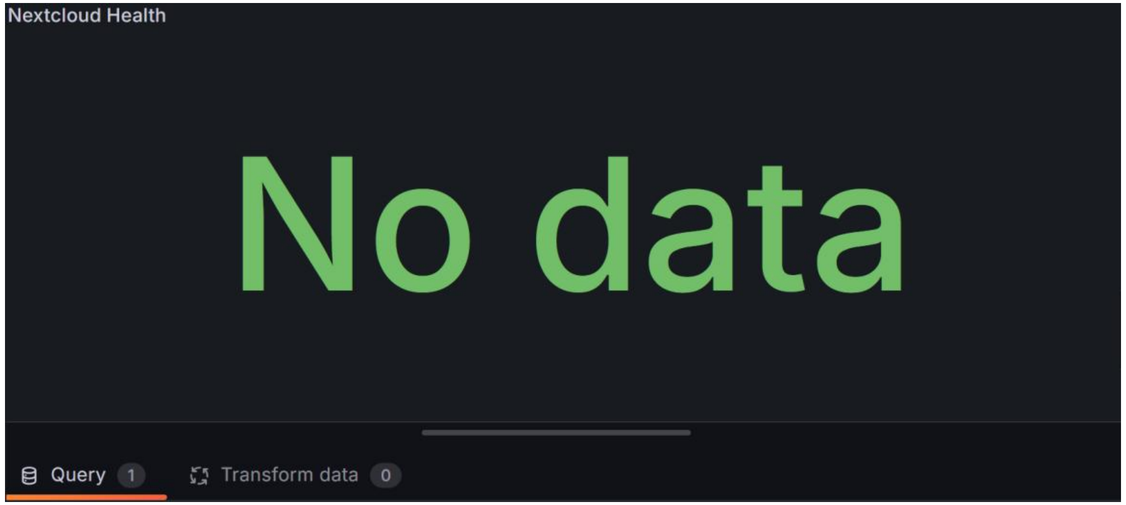
LOAD-TESTING RESULTS: 50MB





CHALLENGES & SOLUTIONS

- The **Nextcloud Exporter** was unable to pull metrics from Nextcloud.
  - Nextcloud was blocking the exporter because it was scraping too frequently.
  - This resulted in: nextcloud_up = 0
- WSL files from the **stress-test** almost **filled up the disk** with thousands of 50MB temporary files
  - deleting them didn't help!
  - The Windows diskpart tool saved the day
- *Many small issues… but I didn't give up!*




```
C:\Users>quit
"quit" non è riconosciuto come comando interno o esterno,
 un programma eseguibile o un file batch.
```

ACHIEVEMENTS

- ✓ **Fully containerized Nextcloud deployment**
- ✓ **Complete monitoring stack** (Prometheus + Grafana)
- ✓ Automated **load testing** with Locust
- ✓ Comprehensive **documentation** and **execution logs**

PERFORMANCE INSIGHTS

- Medium files (5MB):
  median response ~44–49 ms at 10 concurrent users
- System **handles well** moderate concurrent load
- Load testing shows that performance is consistent for any small number of users

SCALING AND OTHER IMPROVEMENTS

- **Scalability**: PostgreSQL + Redis backend; horizontal scaling
- **Security**: Add TLS/SSL via reverse proxy (Traefik/Nginx)
- **Cloud Deployment**: AWS EC2 + S3 with auto-scaling
- **Larger Scale Tests**: GB-sized files and 100+ concurrent users

COST-EFFICIENCY

- All **open-source** components (zero licensing)
- **Lightweight** containers optimize resource usage
- Cloud deployment would use spot instances for cost optimization

THANK YOU